

Robot Specifications

Specs

Parameter	Value
Payload	1 kg near the base, 0.5 kg across the full workspace
Weight	5.5 kg
Reach	400 mm with the standard gripper
Degrees of freedom	6 rotating joints
Material	3D-printed PETG plastic
Power consumption	40 W
Repeatability	0.1 mm
Motors	Stepper motors
Gearboxes	Precision planetary and belt
Position sensing	Limit switches (open-loop version)
Drivers	Open-loop stepper drivers (open-loop version)
Isolated outputs	2
Active CAN buses	1
PC communication	USB
Pneumatic connectors	2

Rotation range

Joint	Range
Joint 1	250°
Joint 2	141°
Joint 3	180°
Joint 4	212°
Joint 5	180°
Joint 6	∞

Operating temperatures

Stepper motors can operate at up to 100–110 °C without issues. For PAROL6 this is not acceptable, since the robot is built from plastic. PETG is recommended because of its high glass transition temperature.

Joint	Holding position (5 h)	Moving (2 h)
Joint 1	51 °C	64 °C
Joint 2	54 °C	64 °C
Joint 3	48 °C	52 °C
Joint 4	60 °C	73 °C
Joint 5	60 °C	72 °C
Joint 6	61 °C	65 °C

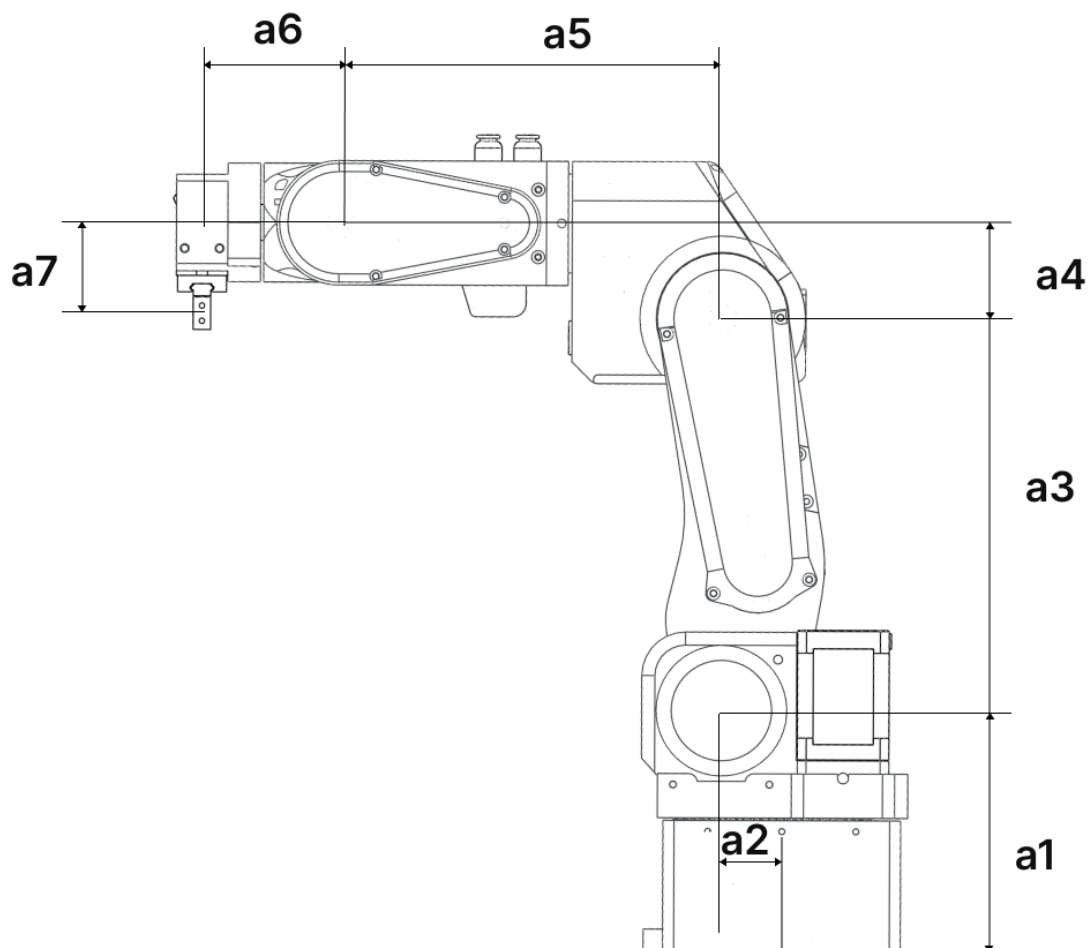
Currents can be adjusted in the PAROL6 control board. The values to adjust are located in `constants.h`. `MOTORx_MAX_CURRENT` sets the maximum stepper current. In `motor_init.cpp`

you can adjust `Joint__->hold_multiplier` where `Hold_current = MAX_current * hold_multiplier`. Reducing currents will lower motor torque and in turn reduce maximum speeds and accelerations.

Danger

When using the robot for longer periods, you **must reduce the current** in the software or you risk destroying your robot.

Dimensions



Dimensions using the default pneumatic gripper.

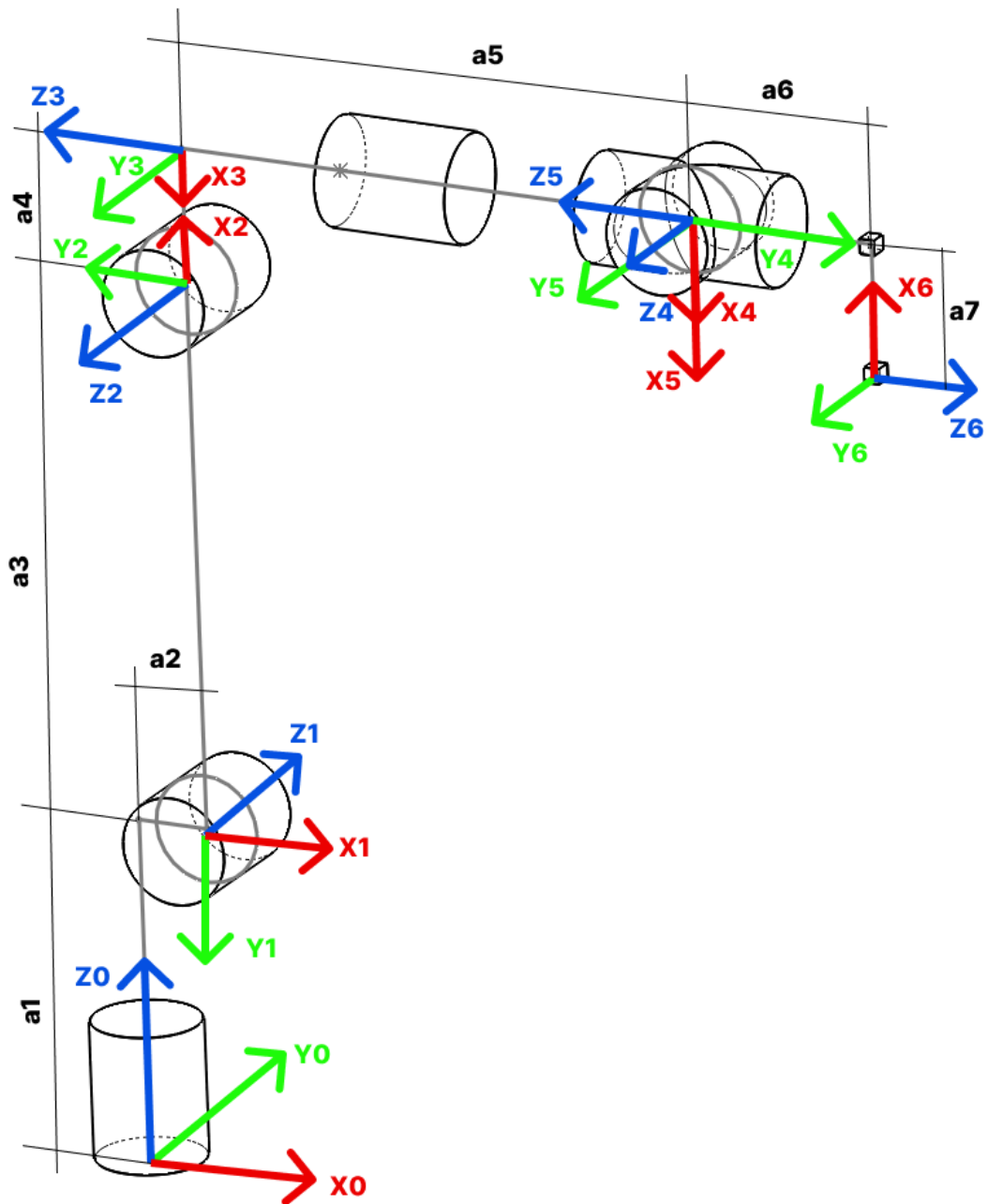
Dimension	Value
a1	110.50 mm
a2	23.42 mm
a3	180.00 mm
a4	43.50 mm
a5	176.35 mm
a6	62.8 mm
a7	45.25 mm

Kinematic diagram



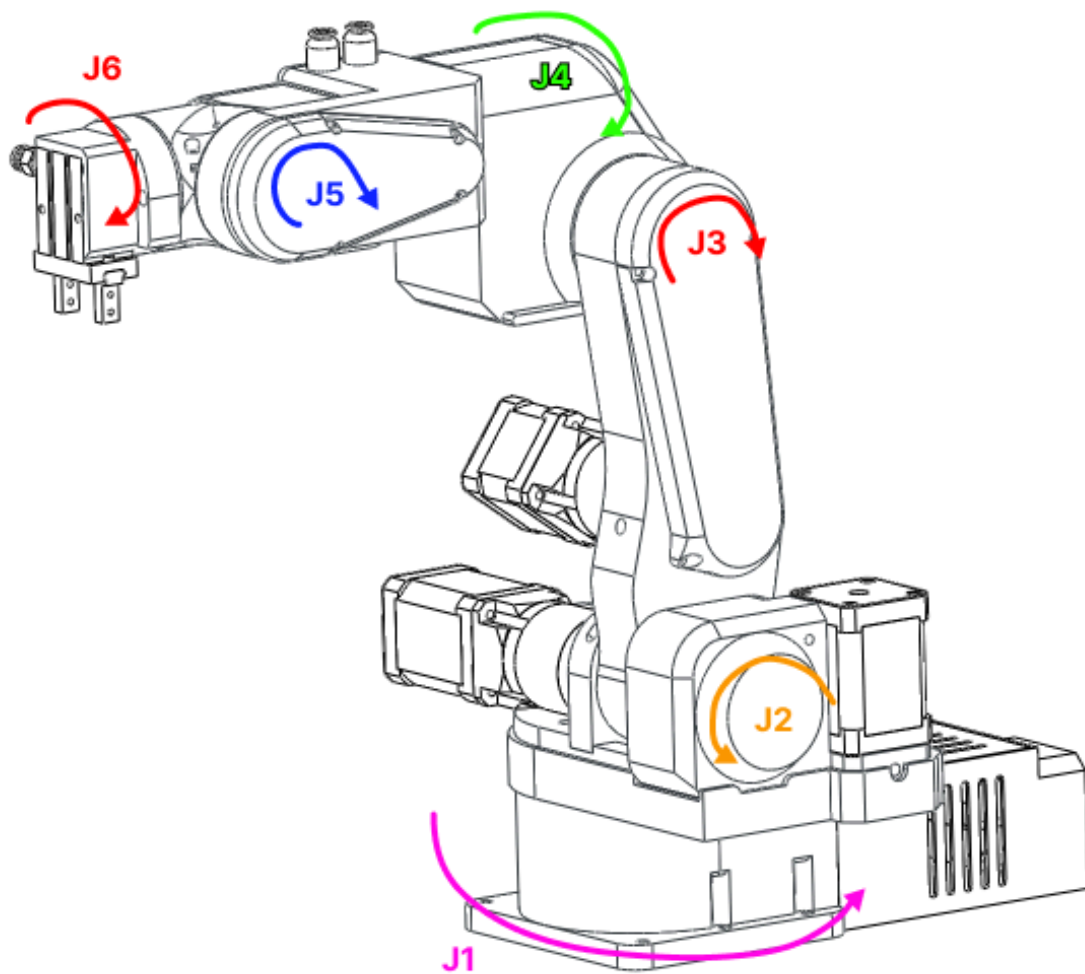
Dimensions can change!

For example, when you change grippers or place the robot on an additional base, you need to update your parameters in the DH table — otherwise your kinematic diagram will be incorrect.



- Kinematic diagram for the robot using the standard pneumatic gripper

Denavit-Hartenberg parameters



Standby position

This is the robot position defined by the DH table below. This position is also called the **standby position**. In this position, joint angles are as follows:

- Joint 1 $\rightarrow 0^\circ$
- Joint 2 $\rightarrow -90^\circ$
- Joint 3 $\rightarrow 180^\circ$
- Joint 4 $\rightarrow 0^\circ$
- Joint 5 $\rightarrow 0^\circ$
- Joint 6 $\rightarrow 180^\circ$

Robot name		CM6			
		Joint angle	Link twist	Link offset	Link len
JOINT	LINK(n,i)	θ THETA	α ALPHA	r_a [mm]	d [mm]
1	Base	TH1	$-\pi/2$	a2	a1
2	Shoulder	TH2 - $\pi/2$	Pi	a3	0
3	Elbow	TH3 + π	$\pi/2$	-a4	0
4	Wrist 1	TH4	$-\pi/2$	0	-a5
5	Wrist 2	TH5	$\pi/2$	0	0
6	Wrist 3	TH6 + π	Pi	-a7	-a6

Joint reduction ratios and microstepping

Reduction ratios for each joint are as follows:

Joint	Type	Ratio
Joint 1	Belt	6.4 : 1
Joint 2	Planetary gearbox	20 : 1
Joint 3	Planetary gearbox $\times$ Belt (38:42)	18.0952381 : 1
Joint 4	Belt	4 : 1
Joint 5	Belt	4 : 1
Joint 6	Planetary gearbox	10 : 1

PAROL6 uses stepper motors with microstepping set to 32 on all motors. With 32 microstepping, a standard 200-step-per-revolution stepper motor requires 6400 steps for one full revolution.

Smallest theoretical step sizes at joint level with 32 microstepping (values are after reduction ratios):

Joint	Step size (rad)	Step size (deg)
Joint 1	0.00015339807878856412	0.0087890625
Joint 2	4.9087385212340514e-05	0.0028125
Joint 3	5.4254478392586896e-05	0.0031085526315789477
Joint 4	0.0002454369260617026	0.0140625
Joint 5	0.0002454369260617026	0.0140625
Joint 6	9.817477042468103e-05	0.005625

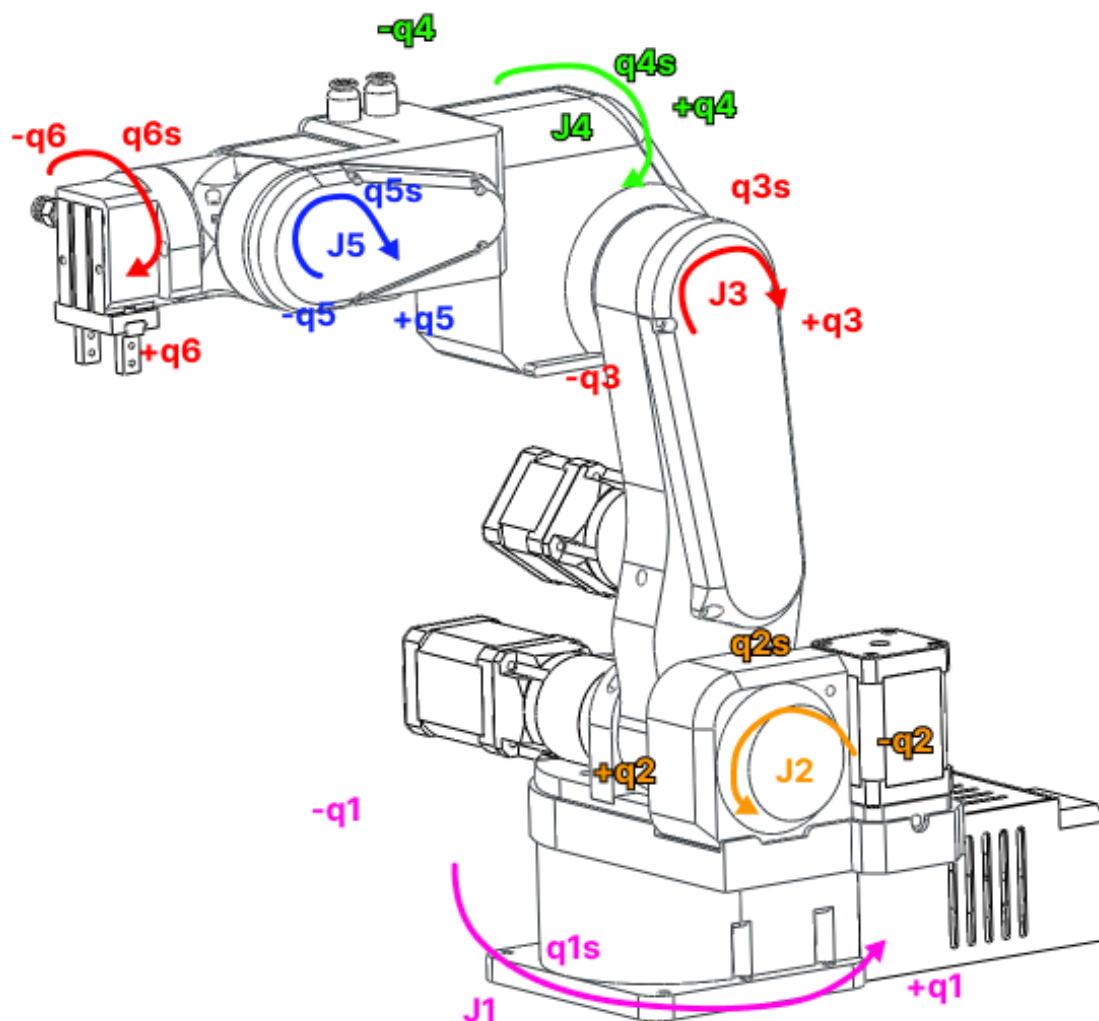
Joint limits



Tip

Joint limits can change depending on the type of gripper or base. When using the robot, make sure you use the correct joint limits for your application.

Robot joint positive rotations are in the directions shown in the image below.



 Note

Values are in degrees.

Joint	Limit in negative direction	Standby position	Limit in positive direction
J1	-123.046875	0	123.046875
J2	145.0088	-90	-3.375
J3	107.866	180	287.8675
J4	-105.46975	0	105.46975

Joint	Limit in negative direction	Standby position	Limit in positive direction
J5	-90	0	90
J6	0	180	360